

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	17/07/2026

Code of Conduct

I N. Sathwik (19A505077) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Sathwik

Instructions

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

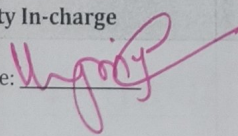
Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:			100 Marks

21	192511330	MARY SHARLIN M	function to print the same in reverse order using stack data structure. 1. Write an algorithm called Find-Largest that finds the largest number in an array using a divide-and-conquer strategy. 2. Analyze how to print all the prime integers present in the Single linked list having a list of 24->21->57->31->12->47->7->38->.
22	192511222	MAYOUN G	1. Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(), Pop(). Print the peak element of the data structure and its index position. With a neat sketch explained in detail. 2. How do you implement Queue data structure using Single Linked List? With a neat diagram discussing Enqueue and Dequeue operations using SLL.
23	192525077	NAGELLA SATHWIK	1. Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Dequeue(), Enqueue(100), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail. 2. Implement Binary searching Algorithm for searching an element and its position having the following 10 elements. Elements are 66, 111, 212, 323, 432, 543, 643, 649, 777, 888. Case 1: Key element = 111, Case 2: Key element=888, Case 3: Key element = 999. Analyse the concept in detail.
24	192525346	NAKKA ADINARAYANA	1. Pictorially insert an element at 3rd position in a doubly linked list, having 6 elements in the list and delete the element at 2nd position from the list. Implement and analyse it. 2. Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Now perform the following operations: 1) Invert the elements in the stack, 2) Pop(), 3) Pop(), 3) Pop(), 4) Push(90), 5) Push(36), 6) Push(11), 7) Push(88), 8) Pop(), 9) Pop(). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
25	192511028	NILAVAN K	1. Convert the following infix expression into a post fix expression. $B - (C/D + (E \% F * G) / H) * J$. Also, evaluate the given postfix expressions. a) 9 3 8 4 /*+ b) 6 2 + 6 3 2. Illustrate the queue operation using following function calls of size = 5. Enqueue(21), Enqueue(67), Enqueue(89), Dequeue(), Enqueue(15), Enqueue(30), Enqueue(52), Dequeue(), Dequeue(), Dequeue(), Dequeue().

38	192511310	VAIBHAVAK LAKSHMI K	- Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. - A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192524458	VASANTHAN S	- Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. - A queue implemented using an array report "Queue Full" even though some positions are empty. - Analyze why this occurs. - Explain how a circular queue solves this problem.
40	192524083	YARRAPPAREDDYG ARI SAI GOPAL REDDY	- Consider the singly linked list: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50$. Delete the node containing 30. Explain how the links change after deletion. - Given two sorted linked lists: L1: $1 \rightarrow 3 \rightarrow 5 \rightarrow 7$ L2: $2 \rightarrow 4 \rightarrow 6 \rightarrow 8$

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows <u>clear</u> understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects <u>appropriate</u> structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

(1)
(0) perform the following operations in circular queue having a size of 5. enqueue(20), enqueue(10), enqueue(30), dequeue(), enqueue(80), dequeue(), enqueue(90), enqueue(100), dequeue(), dequeue(), enqueue(60), enqueue(90). Finally display the front and rear element & with it's position discuss in details.

Ans

Given:-

circular queue of size = 5.

Step	operation	Front	Rear	Queue status
1.	enqueue(20)	0	0	20
2.	enqueue(10)	0	1	20, 10
3.	enqueue(30)	0	2	20, 10, 30
4.	dequeue()	1	2	10, 30
5.	enqueue(80)	1	3	10, 30, 80
6.	dequeue()	2	3	30, 80

Step	Operation	Front	Rear	Queue Status
7.	Enqueue(90)	2	4	30, 80, 90.
8.	Enqueue(100)	2	0	30, 80, 90 100.
9.	Dequeue[]	3	0	80, 90, 100.
10.	Dequeue[]	4	0	90, 100.
11.	Enqueue(60)	4	1	90, 100, 60.
12.	Enqueue(90)	4	2	90, 100, 60, 90

Final queue:

Index	0	1	2	3	4
Queue	100	60	90	-	90

Front position = 4.

Front element = 90.

Rear position = 2

Rear element = 90.

RESULT:

Front = 90 [position]

Rear = 90 [position]

Implement Binary searching Algorithm.
for searching an element and its position.
having the following 10 elements element
are 66, 111, 212, 323, 432, 543, 643, 649, 777,
888.

case :- key element = 111, case2 :- key element
= 888, case3 :- key element = 999 Analyse the
concept in detail.

Ans

ALGORITHM : BINARY SEARCH :-

1. Set $low = 0$, $high = n - 1$
2. Find $mid = (low + high) / 2$.
3. If $key = A[mid]$, return the position.
4. If $key < A[mid]$, set $high = mid - 1$.
5. If $key > A[mid]$, set $low = mid + 1$.
6. Repeat until $low > high$
7. If not found, display element NOT found.

Index	0	1	2	3	4	5
element	66	111	212	323	432	543

case 1: key = 111.

step	low	high	mid	element	RESULT
1	0	9	4	432	key < 432 + High = 3
2	0	3	1	111	FOUND

position = Index 1 (1st position).

case 2: key = 888.

step	low	high	mid	element	RESULT
1	0	9	4	432	key > 432 low = 5
2	5	9	7	649	key > 649 low = 8
3	8	9	8	777	key > 777 low = 9
4	9	9	9	888	FOUND

position = Index 9 (10th position).

case 3:- key = 999.

Step	low	High	mid	Element	RESULT
1	0	9	4	432	key > 432 low = 5
2	5	9	7	649	key > 649 low = 8
3	8	9	8	777	key > 777 low = 9
4	9	9	9	888	key > 888 low = 10

RESULT:-

- case 1:- 111 found at Index 1 (2nd position)
- case 2:- 888 found at Index 9 (10th position)
- case 3:- 999 not found in the array.

Ujjwal